INCH-POUND

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MILITARY SPECIFICATION

GRINDING MACHINE, CYLINDRICAL, EXTERNAL, CENTER TYPE, COMPUTER NUMERICAL CONTROL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric motor driven, floor mounted, horizontal spindle, computer numerically controlled, external, cylindrical, grinding machines with equipment and accessories specified herein.

1.2 Classification. The grinding machine shall be of the sizes listed in table I. The sizes to be furnished shall be as specified (see 3.5 and 6.2).

Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

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(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03a - Gear Classification, Materials and Measuring Methods for Bevel, Hypoid, Fine Pitch Wormgearing and Racks only as Unassembled Gears. (DOD adopted)

AGMA 2000-A88 - Gear Classification and Inspection Handbook. Tolerances and Measuring Methods for Unassembled Spur and Helical Gears (Including Metric Equivalents). (DOD adopted)

AGMA 2001-B88 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth. (DOD adopted)

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts, and Tongues. (DOD adopted)

ANSI B5.9 - Spindle Noses for Toolroom Lathes, Engine Lathes, Turret lathes, and Automatic Lathes. (DOD adopted)

ANSI B5.10 - Machine Tapers. (DOD adopted)

ANSI B7.1 - Safety requirements for the Use, Care, and Protection of Abrasive Wheels. (DOD adopted)

ANSI B11.9 - Safety Requirements for the Construction, Care, and Use of Grinding Machines. (DOD adopted)

ANSI B74.2 - Shapes and Sizes of Grinding Wheels, and Shapes, Sizes, and Identification of Mounted wheels, Specifications

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ANSI B74.13 - Markings for Identifying Grinding Wheels and Other Bonded Abrasives. (DOD adopted)

ANSI/ISO 54 - Cylindrical Gears For General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018-3363.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-232 - Interface Between Data Terminal Equipment and Data Circuit Terminating Equipment, Employing Serial Binary Data Interchange. (DOD adopted)

RS-244 - Character Control for Numerical Machine Control Perforated Tape. (DOD adopted)

RS-267 - Axis and Motion Nomenclature for Numerically Controlled Machines. (DOD adopted)

RS-274 - Interchangeable Variable Block Data Format for Positioning , Contouring, and Contouring/Positioning Numerically Controlled Machines. (DOD adopted)

RS-358 - Subset of American National Standard Code for Information Interchange for Numerical Machine Control Perforated Tape. (DOD adopted)

RS-491 - Interface Between a Numerical Control Unit and Peripheral Equipment Employing Asynchronous Binary Data Interchange Over Circuits Having RS-423-A Electrical Characteristics.

(Application for copies should be addressed to the Electric Industries Association, 2001 I Street, N.W., Washington, DC 20006-1899.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, c/o The National Machine Tool Builders' Association, 7901 Westpark Drive, McLean, VA 22101-4269,)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete grinding machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The grinding machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair. The design and construction of the machine shall be in accordance with either the U.S. Customary System of Units (US) or the International System of Units (SI).

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate machine linear axes position or tool movement shall be graduated in increments not larger than 0.0001 inch for the inch system, not larger than 0.002 mm for the metric system, or not larger than 0.0002 inch and 0.005 mm for the dual system of measurements. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt or remanufactured products are allowed under this specification.

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3.2.3 Energy efficiency. The grinding machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Operating controls. All operating controls necessary for the operation of the machine shall be mounted on either a control console or pendant stand clearly identified to facilitate its use. Voltage and current meters, indicator lights and all other operational control aids shall be arranged and mounted convenient to and in full view of the operator. An emergency stop switch shall be furnished on the control station.

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the grinding machine whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI B11.9. If a conflict arises between 29 CFR 1910 and ANSI B11.9, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration, the procuring activity shall provide part design data to the supplier of the machine (see 6.2). The supplier of the machine shall provide the special guarding which shall meet OSHA 29 CFR 1910.

3.2.6 Mercury restriction. The grinding machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. During operation, service, transportation, or storage of the machine, it shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. The cathode ray tube (CRT) shall display a lubrication fault including a low lubricant condition. An automatic lubrication system shall be provided for all ways and axis screws. Each lubricant reservoir shall have at least an 80-hour capacity and means for determining fluid level. All oil holes, grease fittings, and filler caps shall be readily accessible.

3.2.10 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts except for hand-fitted parts, such as ways, shall be interchangeable without requiring modification. Tooling and accessories normally supplied as standard shall also be considered replacement parts. Replacement parts shall be domestically available.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion or failure.

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3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only when specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the machine shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Drive train gears. All gears of spindle and axis drive trains shall conform to or exceed all provisions of AGMA 390.03a, AGMA 2000-A88 and AGMA 2001-B88 for the inch-pound system of measurement or ANSI/ISO 54 for the metric system of measurement. The gears shall be manufactured of a suitable ferrous material. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The grinding machine shall consist basically of the following components. Any additional components necessary for the operation of the machine shall be furnished in accordance with the manufacturer's standard practice.

3.4.1 Bed and base. The bed and base shall be of a material of sufficient strength and rigidity for supporting and maintaining mutual component alignment as necessary for meeting performance, accuracy and alignment requirements herein. The bed shall have guideways for the support and precision movement of the traversing table along the longitudinal (Z) axis and the grinding wheel base assembly along the cross (X) axis. Worktable guideways shall be of such length that the center of gravity of the headstock and footstock is over the bed at all times during normal grinding operations. Leveling screws and floor mounts shall be provided around the periphery of the base. The hydraulic and coolant reservoirs may be

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housed in the base or provided as separate external units. When applicable, an access door(s) or removable door(s) shall be provided for servicing components housed within the base.

3.4.2 Traversing table. The traversing table shall be a one-piece casting designed to support a swivel table with its components and shall traverse longitudinally (Z axis) on guideways provided by the bed. The table shall have means for preventing over-travel of the table. All table positioning, travel rates and dwell shall be accomplished by numerical control.

3.4.3 Swivel table. The swivel table shall be mounted on the traversing table and shall be capable of swiveling in the horizontal plane for grinding included angles and tapers. A scale, indicating taper, graduated in inches per foot and degrees of angle shall be provided. Means shall be provided for clamping the swivel table in the desired position. When T-slots are provided for longitudinal adjustment of the headstock and footstock, they shall conform to ANSI B5.1.

3.4.4 Grinding wheelhead assembly. The grinding wheelhead assembly shall include a wheelhead housing, spindle and spindle driving mechanism. The assembly shall be mounted on the guideways of the bed and shall traverse cross axially (X axis) to the table. The assembly shall operate at a speed necessary to provide a grinding wheel speed within the range of 5000 to 6500 surface feet per minute. The grinding wheel spindle assembly shall be dynamically balanced and shall be supported by bearings to minimize axial and radial play. The spindle bearings shall be precision antifriction bearings with continuous lubrication or permanently lubricated and sealed for life. The spindle shall be constructed to accept a driving mechanism at one end and a grinding wheel at the other end.

3.4.5 Workhead. The workhead shall be longitudinally adjustable along the swivel table. Means for positive clamping of the workhead to the table shall be provided. Unless otherwise specified (see 6.2), the workhead shall be constructed for dead center grinding. A drive plate complete with drive pin, which revolves around the work center shall be provided. When a live spindle workhead is required (see 6.2), the spindle shall be constructed to receive a driving plate and work center at one end and a driving mechanism at the other. The spindle nose for live and dead center workheads shall conform to ANSI B5.9, and spindle nose taper to ANSI B5.10, designed to receive a work center. Unless otherwise specified (see 6.2), taper number and type shall be the manufacturer's standard. The spindle shall be mounted in precision antifriction bearings with continuous lubrication or permanently lubricated and sealed for life.

3.4.6 Spindle Drive system. Unless otherwise specified (see 6.2), the spindle drive system for both the workhead and the grinding wheelhead assembly shall be powered by an electric motor with belt and pulley arrangement. Means shall be provided for adjusting belt tension unless belt tension is maintained automatically.

3.4.7 Footstock. The footstock shall be longitudinally adjustable along the swivel table. Means for clamping after adjustment shall be provided. The footstock spindle shall be constructed to retract and extend in a spindle mount. A handwheel, lever or other means shall be provided for adjusting spindle movement. Unless otherwise specified (see 6.2), the footstock spindle shall have an internal taper identical in number and type with that of the workhead spindle. The spring tension of the footstock center shall be adjustable to ensure

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proper pressure for the weight and size of the workpiece. The footstock shall have a manual adjustment with a graduated dial or gauge to facilitate taper correction. All moving parts shall be shielded against abrasive grit and coolant.

3.4.8 Wheel truing device. Unless otherwise specified (see 6.2), a wheel truing device, complete with not less than one industrial diamond, shall be rigidly mounted on the machine and shall be compatible with the programmable wheel truing cycles described herein. Programmable wheel truing cycles shall include dressing at cycle start, a single interruption at rough grinding cycle completion, or both; multiple dress; or in cycle dress by manual data input (MDI). The wheel truing device shall be capable of dressing basic or complex linear and circular forms utilizing the machine's own X and Z axes.

3.4.9 Grinding wheels. Unless otherwise specified (see 6.2), grinding wheels shall conform to ANSI B74.2. The wheels shall be marked in accordance with ANSI B74.13 and shall meet the safety requirements established in ANSI B7.1. All wheels shall be properly guarded.

3.4.10 Computer numerical control. The control shall be completely programmable through a fully automatic, solid-state, soft-wired, integrated circuit type computer numerical control. The control shall accept and translate program data from memory into positioning and feed motion command signals for the X and Z axes. Spindle speed, direction and path of axes movements, and sequence and timing of all operations shall be directed by the control. The control shall be capable of linear and circular interpolation. Axes (X and Z), spindle, and motion nomenclature shall conform to EIA RS-267. Control features shall include fixed cycles, control diagnostics, part program and buffer edit capability, and part program storage. The machine control unit (MCU) shall accept part program data entered from a keyboard MDI and from an external input device. The control shall be provided with both EIA Standards RS-232 and RS-244 interface plugs for data input and output devices. A line voltage of $\pm$10 percent from normal shall not adversely affect the computer numerical control function. The control shall be capable of functioning in ambient temperatures ranging from 50^{0} to 120^{0} Fahrenheit (F) and relative humidities ranging from 5 to 95 percent noncondensing. The control shall automatically shut down when internal temperature exceeds safe operating temperature. The control resolution shall be no greater than 0.0001 inch.

3.4.10.1 Data display. A CRT capable of displaying at least 250 characters simultaneously shall be provided. The control shall display the active and a minimum of the next two lines of information both in single block and automatic modes. The screen shall be automatically updated as each block of data is executed. Unless otherwise specified (see 6.2), the CRT shall, as a minimum, display the following data:

a. Mode of operation (i.e. auto., single block, MDI, and manual).

b. Machine status (i.e. power on, in cycle, idle, end of cycle, in hold, in E-stop).

c. Control status (i.e. manual spindle override, manual feedrate override, in edit, program transfer active).

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d. Actual and command position of each axis.

e. Program listing.

f. Subroutine index.

g. Tool offset.

h. Control and system error messages.

3.4.10.2 External data input. The master control unit (MCU) shall accept part program data from an external input device compatible with the part program storage media (magnetic cassette, bubble cassette, diskette, magnetic card or punched tape) of the type specified (see 6.2). If punched tape is required the input and codes shall comply with EIA Standards RS-227, RS-274 and RS-358.

3.4.10.3 Switchable inch/metric input. The control shall provide switchable inch/metric input capability.

3.4.10.4 Part program memory. Unless otherwise specified (see 6.2), the MCU shall be capable of storing in solid-state or bubble-type memory the content of not less than 256k bytes of user accessible main memory.

3.4.10.5 Standby battery power. Unless otherwise specified (see 6.2), standby battery power shall be provided for machines having volatile memory. The standby battery power shall be capable of retaining a fully loaded memory for a period of not less than 72 hours after the primary power source fails.

3.4.10.6 Part program edit. A program edit capability shall be provided to allow block insertions, additions, deletions or modifications of a part program in the part program memory. Data related to part program edit shall be input by manual data input provided on the control panel. Edited program memory shall be automatically acted upon, in the sequence designated, whenever the program is cycled.

3.4.10.7 Constant surface speed. The constant surface grinding speed control shall permit maintaining a constant grinding speed rate, within the limits of the machine, regardless of part diameter. Wheelhead spindle speed control shall be based on the part diameter and the programmed surface speed to maintain a constant grinding rate.

3.4.10.8 Feed rate override. A feed rate override control shall be provided to allow manual adjustment of the programmed feed rate from 10 to not less than 120 percent of the programmed rate in either continuous adjustment or in steps of not more than 10 percent. The feed rate override control shall have a position that allows return of feed rate to that originally programmed.

3.4.10.9 Jog. The jog control shall permit the movement of the slide in the plus or minus direction of travel at preset jog rates such as high, low, or selectable rates of jog. Jog control shall allow the operator to move the grinding wheel away from the workpiece to inspect grinding wheel and workpiece size during any grinding program and return to the correct position automatically.

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3.4.10.10 Sequence number search. The control shall have the capability to permit the operator to search through the part program data for a designated block number by initiating the search through the use of the MDI keyboard. The search may be conducted through the tape data or stored part program data, depending on which is active.

3.4.10.11 Block delete. The control shall have a delete feature to permit bypassing one or more instructions of programmed data.

3.4.10.12 Program data edit. Part program edit capability shall be provided as means for editing tape and MDI input to optimize part program data. It shall be possible to modify part program data by adding, deleting, correcting, or completely changing stored data and operating with the edited program.

3.4.10.13 Diagnostic capability. The control shall include a diagnostic system for monitoring the control and grinding machine during operation and for locating malfunctions when troubleshooting. It shall bring the system to a controlled stop if a malfunction occurs during operation. The system shall give fault data in understandable code or message form on the display unit which will identify a problem at board level in the control and identify program errors. Unless otherwise specified (see 6.2), as a minimum the diagnostic capability of the control shall determine and display the following fault conditions.

a. Guard door open.
b. Illegal program entry.
c. Axis feedback loop fail.
d. Axis travel limits exceeded.
e. Low hydraulic pressure.
f. Axis servo motor failure.
g. Control console overheat.
h. Lubrication fault.
i. Excessive wheel wear.
j. Program error.
k. Feed limit.

3.4.10.14 Control system modes of operation. The operational modes of the control shall include manual, automatic, single block and MDI.

3.4.10.15 Automatic cycle. Unless otherwise specified (see 6.2), the control shall have the capability of running in automatic cycle. Once programmed the control shall automatically run for an infinite number of parts.

3.4.10.16 Manual data input. A means for manual data input shall be furnished to permit manual entry of alphabetic, numeric, and special characters necessary for control of all programmable machine functions in both EIA Standards RS-491 and RS-274 format. The manually input data shall be immediately displayed on the data display to permit data verification. The MDI shall permit entering a block of data into memory upon command.

3.4.10.17 Canned cycles. Canned cycles shall be provided as specified and fully described (see 6.2)

3.4.10.18 In-process gaging. When specified (see 6.2), in-process gaging shall be provided as required and fully described by the procuring activity.

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3.4.10.19 Computer graphics. When specified (see 6.2), computer graphics capabilities shall be provided and fully described by the procuring activity.

3.4.10.20 Conversational programming. When specified (see 6.2), conversational programming shall be provided as required and fully described.

3.4.10.21 Postprocessor. When specified (see 6.2), the system shall be furnished with a postprocessor program(s) to be utilized in off-line computer-assisted preparation of part programs. The postprocessor shall convert specified input information to the form and format required to operate the machine tool and control. The postprocessor shall be written for use on the type computer specified.

3.4.11 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit a full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or strainer which is easily removable for cleaning. The reservoir shall be equipped with means for emptying and cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant over the entire work area.

3.4.12 Hydraulic system. The hydraulic system shall conform to or exceed the requirements of JIC H-1. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, a provision for draining, and a cleanout access plate if the reservoir is not removable.

3.4.13 Electrical system. The electrical system shall be wired to conform to NFPA 79. Each machine shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), the machine shall be designed for operation on 3-phase, 60-hertz (Hz), dual 230/460 voltage. Unless otherwise specified (see 6.2), the grinding machine shall be initially wired for operation on 460 volts.

3.4.13.1 Motors. Motors shall meet the requirements of 3.4.13. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating-current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.13.2 Worklight. Unless otherwise specified (see 6.2), a worklight conforming to 3.4.13 shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.13.3 Hour meter. When specified (see 6.2), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time for the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than

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1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the grinding machine shall meet the requirements of table I for the machine specified. Machine size and capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.6 Alignment tolerances. Each machine shall meet the alignment tolerances specified in table II.

3.7 Performance. The grinding machine shall be capable of grinding workpieces in accordance with the test requirements in 4.8.

3.7.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.8 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2).

3.10 Repair parts. Required repair parts shall be furnished as specified and fully described (see 6.2).

3.11 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.11.1 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the plate shall be as listed below.

Points of lubricant application
Type of lubricant
Servicing interval
Military or Federal specification for each lubricant (if applicable)

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating

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Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11.3 Hydraulic fluid requirements plate. When specified (see 6.2), a hydraulic fluid requirements plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2), the plate shall contain the following information.

Type of fluid (SAE or military/Federal specification number)
Type of filter

3.12 Workmanship. Workmanship of the grinding machine and its accessories shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

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4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. Acceptance test shall be performed on each grinding machine to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. The machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41 to 104° F and 20 to 95 percent relative humidity.

4.8.2 Operational test. Each grinding machine shall be operated under CNC for a period of 4 hours without load. The CNC program shall cycle each axis through its entire range, addressing all speeds, feeds, direction of rotation, and standard control functions specified in this document. Proper operation of all machine functions, controls, motors, adjusting mechanisms, and accessories shall be verified during the test period.

4.8.3 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.6. The alignment tolerance test shall be performed prior to and subsequent to the performance test given in 4.8.4 through 4.8.4.1. Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

4.8.4 Performance tests. The test part shall be as shown in figure 1. "A" diameter and "B" length shall be not less than one-half of the capacity for the machine being tested. All other dimensions and workpiece material shall be at the contractor's option. Utilizing MDI programming, the contractor shall demonstrate the machine's capability to step grind (at least 2 diameters), traverse grind and face grind. All diameters shall be round within 0.0001 inch of a desired dimension with not less than 0.0001 inch taper per foot and a surface roughness not more than 5 microinches. Axial runout of the ground shoulder shall not exceed 0.0001 inch total indicator reading (TIR). Single point contouring of radius "C" and angle "D" shall be accomplished utilizing circular and linear interpolation. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

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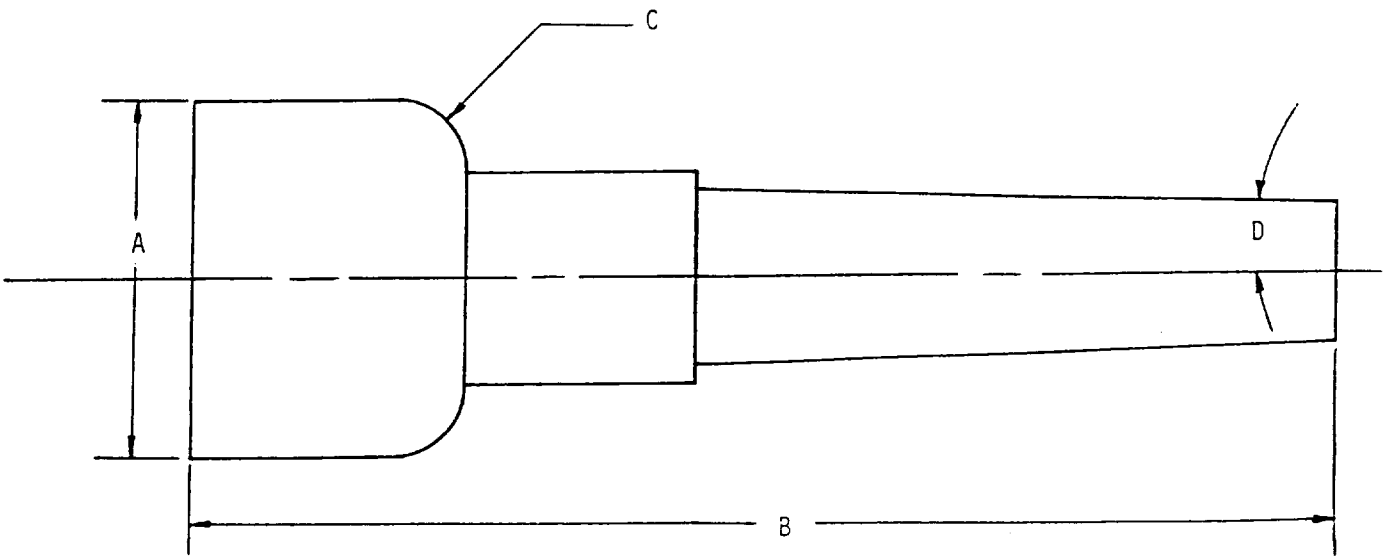


Figure 1. Cylindrical performance test part

4.8.4.1 Additional performance tests. Performance tests required in addition to or in lieu of the requirements given in 4.8.4 shall be as specified and fully described (see 6.2).

4.8.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Grinding machines described in this document are intended for use in a production environment where multiple parts of the same configuration are to be ground from part programs input to the CNC by MDI or an external input medium. The grinders are designed for use where very close tolerances are required for outside diameter straight, step, taper, and face grinding.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

j. Workhead center, if different (see 3.4.5)

k. Specify if a live spindle workhead is required (see 3.4.5).

l. Specify workhead spindle taper number and type, if different (see 3.4.5)

m. Spindle drive systems, if different (see 3.4.6).

n. Footstock spindle taper number and type, if different (see 3.4.7).

o. Wheel truing device, if not required (see 3.4.8).

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p. Grinding wheels, if different (see 3.9).

s. Data display, if different (see 3.4.10.1).

t. External data input media, specify and fully describe (see 3.4.10.2).

u. Part program memory, if different (see 3.4.10.4).

v. Standby battery power, if different (see 3.4.10.5).

w. Diagnostic features, if different (see 3.4.10.13).

x. Automatic cycle, if different (see 3.4.10.15).

y. Specify canned cycles, if required (see 3.4.10.17).

z. Specify and fully describe in-process gaging, if required (see 3.4.10.18).

aa. Specify and fully describe computer graphics, if required (see 3.4.10.19).

bb. Specify and fully describe conversational programming, if required (see 3.4.10.20)

cc. Postprocessor, if required (see 3.4.10.21).

dd. Electrical system, if different (see 3.4.13).

ee. Initial voltage, if different (see 3.4.13).

ff. Motors, if different (see 3.4.13.1).

gg. Worklight, if not required (see 3.4.13.2).

hh. Hour meter, if required (see 3.4.13.3).

ii. Size and capacity, if different (see 3.5).

jj. Additional performance features, if required (see 3.7.1).

kk. Specify and fully describe optional equipment, if required (see 3.9).

ll. Specify applicable repair parts, if required (see 3.10).

mm. Lubrication plate, if required (see 3.11.1).

nn. Information to be provided on lubrication plate, if different (see 3.11.1).

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oo. Nameplate, if different (see 3.11.2).

pp. Hydraulic plate, if required (see 3.11.3).

qq. Hydraulic plate, if different (see 3.11.3).

rr. Inspection conditions, if different (see 4.3).

ss. First article inspection, if different (see 4.4).

tt. Quality conformance inspection, if different (see 4.5).

uu. Acceptance test, if different (see 4.6).

vv. Additional performance tests, if required (see 4.8.4.1).

6.3 Technical data. All technical data ordered using a DD Form 1423 should be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le System International D'Units (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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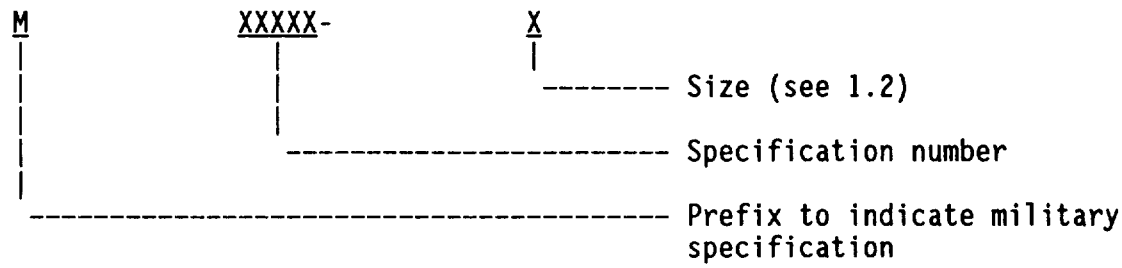
6.8 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.9 Training. Training required by the procuring activity should be provided as specified in the contract.

6.10 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.11 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.12 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



6.13 Subject term (keyword) listing.

Automatic
Cylindrical
External
Plunge
Production
Traverse

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TABLE I - Sizes and capacities style A machines.

Characteristic	Requirement						
Sizes	612	912	1018	1218	1418	1630	2424
Optional sizes - In 6" increments (same as rated swing)	618-648	918-930	1024-1060	1224-1230	1424-1448	1636-1648	2430-2448
In 24" increments (same as rated swing)					1472-14120	1672-16120	2472-24120
Swing over table, inches	6	9	10	12	14	16	24
Distance between centers, inches	12	12	18	18	18	30	24
Optional distances between centers - 6" increments	18-48	18-30	24-60	24-30	24-48	36-48	30-48
24" increments	---	---			72-120	72-120	72-120
Swivel table, amount of swivel angle in degrees							
Toward wheelhead	11	12	12	12	12	12	--
Away from wheelhead	2	3	3	3	5	5	--
Swivel table, graduated taper per foot, inches							
Toward wheelhead	5	5	5	5	5	5	3-3/8
Away from wheelhead	1	1	1	1	2	2	1-1/4
*Grinding wheelhead:							
Grinding wheel (outside diameter by face width by hole diameter, inches)	20x2x12	24x2x12	24x2x12	24x2x12	24x2x12	30x3x12	30x3x12
Spindle speed, RPM	1240	1030	1030	1030	830	830	830
*Headstock:							
Speeds, number	<----------infinite---------->						
Speed range, RPM	97-395	97 - 395	66 - 268	66 to 268	63 to 212	40 to 144	11 to 44
*Table traverse:							
Grinding rates, number	<----------infinite---------->						
Range of grinding rates, inches per minute	3 to 240	3 to 240	3 to 240	3 to 240	3 to 220	3 to 120	3 to 120
Table dwell adjustment, seconds	0 to 30	0 to 30	0 to 30	0 to 30	0 to 30	0 to 30	0 to 30
*Infeed adjustments:							
Infeed increment, diameter reduction, inches, not more than	.00003	.00003	.00003	.00003	.00003	.00003	.00003
Automatic infeed at table reversal, diameter reduction, range, inches	<----------.0001 - .015---------->						
Total adjustment of wheelhead by hand	7-1/2	9-1/2	11-1/2	13-1/2	15-1/2	17-1/2	25-1/2
*Motors, horsepower:							
Wheelhead	7-1/2	7-1/2	7-1/2	7-1/2	15	20	20
Headstock	1/2	1/2	3/4	3/4	1	1-1/2	3
Hydraulic pump	1	1	1	1	1-1/2	1-1/2	2
coolant pump	1/4	1/4	1/4	1/4	1/4	1/4	1/3

* Unless otherwise specified (see 3.5)

NOTE: 1 - Machine sizes and capacities shall not be less than the stated requirements of the size specified in 6.2.1, nor shall they be greater than the next larger size shown in table 1 above. Provided the largest size and capacity is specified, it may not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the minimum or less to the maximum or greater.

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TABLE II. Alignment tolerances.

Tests to be applied	Maximum Allowable Deviation
Bed and table:	
Bed straight in longitudinal direction	0.00025" per ft.
Bed flat or level in transverse direction, no twist permitted	0.00025" per ft.
Guideways of headstock and footstock parallel with table movement	0.00012" per ft.
Wheelhead and spindle:	
Axis of wheelhead parallel with table movement in vertical and horizontal plane	0.0002" per ft.
Wheelhead spindle radial runout	0.0005" TIR
Wheelhead spindle axial slip at two points, displaced by 180 degrees	0.0002" TIR
Wheelhead spindle flange axial runout	0.0002" TIR
Workhead and spindle:	
Axis of workhead parallel with table movement in vertical and horizontal plane	0.0002" per ft.
Workhead spindle radial runout	0.0001" TIR
Workhead spindle axial slip at two points displaced by 180 degrees	0.0001" TIR
Workhead spindle flange axial runout	0.0001" TIR
Footstock:	
Sleeve parallel with table movement in vertical and horizontal plane	0.0005"

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA - GS

User activities:
Navy - AS, OS

Preparing activity:
DLA - IP

Project (3415-0182)

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INSTRUCTIONS

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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-G-80278	900413

3. DOCUMENT TITLE: Grinding Machine, Cylindrical, External, Center Type, Computer Numerical Control

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Industrial Plant Equipment Center ATTN: DIPEC-SSM	(1) Commercial 901-775-4797	(2) AUTOVON 638-4797
c. ADDRESS (Include Zip Code) 2163 Airways Blvd Memphis, TN 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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